Please read the instructions below before the exam starts, and listen carefully to the instructions given by the invigilators.

You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2024

EC114 2023/4 A 800
EC114 2022/3 A 800
EC114 2022/3 A 850

Graph paper
Only Casio FX 83GT CW, Casio FX 85GT CW, Casio FX-85MS calculators are permitted
Dictionaries are not permitted

UNIVERSITY OF READING

INTRODUCTORY MACROECONOMICS (EC114)

Three hours

Answer THREE questions as follows
Section A: Answer ALL questions
Section B: Answer ONE question
Section C: Answer ONE question

Answers should include brief, concise and to the point explanations.
All graphs must be clearly and accurately labelled.

Section A
Answer Question 1

1. International Trade

(a) Briefly explain the following concepts; use graphs and equations if necessary.

(i) Real Exchange Rate

(ii) Heckscher-Ohlin Theory

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) A country's balance of payments is the same as its current account.

(15 marks total)

(b) Consider an economy which is given by the following equations:

$C = 80 + \frac{3}{4}Y_D$ $\quad\quad$ $I = 49$

$G = 60$ $\quad\quad$ $T = 20$

$IM = 0.05Y_D$ $\quad\quad$ $EX = 20$

(i) Calculate the equilibrium level of real GDP for this economy. What is the size of the government deficit? What is the size of the current account balance?

(ii) If government spending increase to 75, what happens to equilibrium output and imports?

(iii) Suppose a quota is imposed which limits imports to 25. How do your answers to parts (i) and (ii) change with this quota?

(35 marks total)

Section B
Answer ONE of the two questions

2. Economic Basics

(a) Briefly explain the following concepts; use graphs and equations if necessary.

(i) Absolute Advantage

(ii) Opportunity Cost

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) Scarcity implies that at the current market price, consumers are willing to purchase more of a good than suppliers are willing to produce.

(15 marks total)

(b) You are given the following information about Kenya and India, which are two countries that produce and consume only rice and tea. Each country has 100 units of labour.

	Rice per unit of labour	Tea per unit of labour
Kenya	4	12
India	6	12

(i) Draw the PPFs for both countries (Putting rice on the horizontal axis). If there is no trade, what is the price of tea in both countries? Explain why the prices are the same/different in the two countries.

(ii) Which country has an absolute advantage in which good? Which country has an comparative advantage in which good? Briefly explain your answer.

(iii) If these two countries decided to trade, which country would export which good? Briefly explain. What can you say about the relative price of rice once trade begins?

(35 marks total)

3. Money and Monetary Policy

(a) Briefly explain the following concepts; use graphs and equations if necessary.

(i) Money Multiplier

(ii) Open Market Operations

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) To increase the money supply, the Fed will purchase government securities.

(15 marks total)

(b) Assume that Fleet Bank is the only bank in the economy of Readingland. No one in Readingland holds any currency (they carry out all transactions using debit cards or credit cards). Fleet Bank currently has £100 million in deposits and faces a reserve requirement of 10%.

(i) Show a T-account for Fleet Bank. Now suppose you win £10 million in the lottery and deposit this entire amount into your current account. What is the immediate impact on reserves for Fleet Bank? How much are excess reserves?

(ii) Suppose that Fleet Bank always lends out any excess reserves. If the reserve requirement is 10%, what will be total change in deposits in this economy? Draw a T-account showing the final balance sheet of Fleet Bank.

(iii) Now suppose that the Bank of England announces that the new reserve requirement is 20%. What action must Fleet Bank take in order to meet the new reserve requirement (explain in words – no need for calculations)? What do you think the Bank of England was trying to achieve when it raised the reserve requirement?

(35 marks total)

Section C
Answer ONE of the two questions

4. Keynesian IS-LM Model

(a) Briefly explain the following concepts; use graphs and equations if necessary.

(i) Budget Deficit

(ii) Planned Aggregate Expenditure

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) Fiscal drag is the result of increasing government spending.

(15 marks total)

(b) Consider an economy where the consumption function is given by $C = 500 + 0.8(Y - T)$. Investment is given by $2000 - 100r$ (where r is the real interest rate), government spending is 5000, taxes are equal to 4000. Exports are 1000, while imports are given by $IM = 0.2Y$.

(i) Suppose that the real interest rate is $r = 5$. Calculate the equilibrium level of real GDP for this economy. Now suppose that the Fed decides to raise the interest rate to $r = 10$. Calculate the new equilibrium GDP for this economy.

(ii) Continue to assume that $r = 10$. If the government decides to raise spending by 1000, what would be the new equilibrium level of GDP? Would you expect this change in GDP to have any impact on the interest rate? Explain why (not).

(iii) Some economists say that consumption also depends on the real rate of interest. If that were the case, explain in words how that would affect your answer in part (ii).

(35 marks total)

5. Labour Markets and Inflation

(a) Briefly explain the following concepts; use graphs and equations if necessary.

(i) Natural Rate of Unemployment

(ii) NAIRU

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) An increase in worker productivity will cause the labour supply curve to shift.

(15 marks total)

(b) Consider a simple economy where labour markets are characterized by the following equations for supply and demand respectively (where W is the real wage):

$$L_S = 10W - 18 \quad \text{and} \quad L_D = 190 - 6W$$

(i) If we have a classical economy, calculate all equilibrium values for this labour market. Graphically illustrate equilibrium. How much unemployment do we have in this economy?

Now suppose instead we have a Keynesian economy. In addition, suppose that the labour force is composed of 120 individuals and that the natural rate of unemployment is 10%

(ii) Suppose the real wage is sticky at a level below the equilibrium real wage from part (i). What can we conclude about the demand for labour? What do we know about the output gap and where we are in the business cycle? Using an AS-AD graph illustrate short-run equilibrium.

(iii) Using the short-run Phillips curve, what can you conclude about inflation? Show this graphically. What does the long-run Phillips indicate will happen to both inflation and unemployment? Illustrate this graphically.

(35 marks total)

(End of question paper)